

CONTEXTDRIVE

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Bachelor of Technology

by

Angith Vijayan Ne (VML23CS046)

Christo Santhosh (VML23CS082)

Sebastian Mathew Parayidathil (LVML23CS250)

Umarsinan N (VML23CS234)

under the supervision of

Ms. Anju Sreedharan

Assistant Professor



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VIMAL JYOTHI ENGINEERING COLLEGE CHEMPERI

CHEMPERI P.O. - 670632, KANNUR, KERALA, INDIA

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DEPT. OF COMPUTER SCIENCE AND ENGINEERING CERTIFICATE

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PROJECT COORDINATORS

Ms. Dinsha P K

Assistant Professor

Computer Science & Engineering

Vimal Jyothi Engineering College

Dr. Nithya V P

Associate Professor

Computer Science & Engineering

Vimal Jyothi Engineering College

Dr. Renji P Cheriyan

Professor

Computer Science & Engineering

Vimal Jyothi Engineering College

PROJECT GUIDE

Ms. Anju Sreedharan

Assistant Professor

Computer Science & Engineering

Vimal Jyothi Engineering College

Place : Chemperi

Head of the Department

Date : 05-10-2026

(Office Seal)

DECLARATION

We hereby declare that the project report **CONTEXTDRIVE**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bona fide work done by us under the supervision of **Ms. Anju Sreedharan**.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma or similar title of any other University.

CHEMPERI
05-10-2026

Angith Vijayan Ne
Christo Santhosh
Sebastian Mathew Parayidathil
Umarsinan N

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Angith Vijayan Ne
Christo Santhosh
Sebastian Mathew Parayidathil
Umarsinan N

Abstract

Road safety is significantly affected by a combination of factors, including driver inattention, adverse weather conditions, vehicle speed, and complex traffic environments. This project, **ContextDrive**, presents a smartphone-based, context-aware driver assistance system designed to improve road safety by assessing driving risk in real time.

The system utilizes a standard smartphone equipped with a rear camera, GPS receiver, and internet connectivity to gather comprehensive driving data. It integrates real-time camera frames for object detection, GPS data for speed and location analysis, weather information via the OpenWeatherMap API, and time-based context. A lightweight object detection model, YOLOv8 Nano (YOLOv8n) running on Google LiteRT (formerly TensorFlow Lite), identifies vehicles and obstacles, while an intelligent engine analyzes these multiple driving factors together rather than treating individual hazards independently.

To ensure transparency and build driver trust, ContextDrive provides transparent safety recommendations based on the combined contextual data. The system continuously estimates the current driving risk score and delivers timely alerts to the driver. The proposed architecture performs entirely on-device, ensuring real-time responsiveness and privacy without the need for expensive or specialized external hardware.

By combining computer vision, edge computing, and context-aware risk analysis, the platform provides drivers with an affordable, portable, and practical solution. ContextDrive enhances situational awareness, promotes safer driving habits, and offers an intelligent approach to proactive accident prevention.

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List of Abbreviations

ADAS	Advanced Driver Assistance Systems
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DFD	Data Flow Diagram
ER	Entity Relationship
GPS	Global Positioning System
JSON	JavaScript Object Notation
LBS	Location-Based Services
ML	Machine Learning
UI	User Interface
YOLO	You Only Look Once

Chapter 1

Introduction

1.1 Overview

Road safety depends on several important factors such as vehicle speed, weather conditions, visibility, traffic density, and driver attention. Drivers often make decisions based on personal experience, traditional practices, or limited observations of their immediate surroundings. This can make safe driving difficult, especially when environmental and traffic conditions change rapidly.

ContextDrive is a smartphone-based context-aware driver assistance system developed to help drivers make safer driving decisions using sensor data and Artificial Intelligence. The system combines computer vision, global positioning systems (GPS), weather intelligence, and transparent decision logic in a single platform.

The system uses a standard smartphone equipped with a rear camera to collect important visual parameters such as nearby vehicles and obstacles. GPS information is also considered for analyzing the current speed and location. A machine learning model uses these inputs to detect road users and estimate the distance to potential hazards.

In addition to object detection, the system provides contextual risk assessment and safety guidance based on the available telemetry and environmental information. A transparent rule-based engine helps the user understand the factors influencing the current risk level. Weather intelligence provides real-time rain and visibility-related information to support better situational awareness.

The system also includes a multi-dimensional context vector that continuously updates the driving risk state. An intuitive and responsive interface makes the platform easy to use directly from a dashboard-mounted mobile device.

1.2 General Background

Modern driving can benefit from technologies that provide accurate and timely information about road conditions. Smartphone sensors allow vehicle telemetry and visual data to be collected directly from the dashboard. In this system, the smartphone camera and GPS receiver gather live readings and make the information available to the application.

Machine learning can be used to analyze visual and environmental parameters and iden-

tify potential hazards on the road. This helps reduce dependence on manual risk estimation by the driver. Explicit reasoning further improves the system by showing the important factors that influence a safety recommendation, ensuring the driver understands why a warning was issued.

Weather conditions also have an important role in driving safety. Visibility, rainfall, and time of day can severely affect braking distance and reaction times. Therefore, the system combines visual, telemetry, and weather information to provide more useful and context-aware driving guidance.

Contextual interactions are also considered because driving risk depends not only on a single hazard but also on how multiple factors combine. The risk engine provides a comprehensive assessment by evaluating speed, proximity, and environment together. Thus, the system brings different driving information sources together in one cohesive platform.

1.3 Problem Statement

Drivers need to make important split-second decisions regarding speed, following distance, and maneuvering. However, the required information is often available through different and disconnected sources or is difficult to perceive manually. Drivers may have to separately judge traffic distance, monitor their speed, and mentally adjust for weather or visibility conditions.

Many existing driver assistance systems mainly focus on detecting a specific hazard in isolation, such as a forward collision or lane departure. Such systems may not provide integrated contextual monitoring, transparent explanations for their warnings, or live environmental adjustments in the same affordable platform. Access to reliable, multi-factor safety information can also be difficult for users who do not own vehicles with expensive, built-in advanced driver assistance systems (ADAS).

1.4 Scope of the System

The ContextDrive system covers several major areas of driver assistance and safety monitoring. It provides real-time object detection using a smartphone camera, GPS-based telemetry insights, weather-aware risk adjustments, and time-based contextual analysis. The system also includes explicit reasoning logic for understanding risk predictions and a multi-dimensional context vector for analyzing combined driving factors. A responsive dashboard interface allows drivers to access these features directly from a mounted mobile device.

The system also provides on-device data processing for ensuring privacy and low-latency

performance without requiring specialized external hardware. It is intended to support drivers in maintaining situational awareness and making safer decisions rather than replacing their attention or active vehicle control.

1.5 Objectives

- To develop a smartphone based context-aware driver assistance system for real time driving risk assessment.
- To integrate camera, GPS, weather, and time based information for comprehensive driving context analysis.
- To generate a risk score and provide timely, transparent safety recommendations.
- To improve driver awareness and promote safer driving without requiring additional hardware.

Chapter 2

Literature Review

2.1 Personalised Driver Risk Assessment with Adaptive Feedback using Crowdsensed Telemetric Data

This paper by Auwal Sagir Muhammad et al., published in IET Intelligent Transport Systems (IET, 2025), presents a comprehensive framework for assessing driver risk using telemetric data. The study focuses on transforming raw GPS traces and temporal data into structured driving risk indicators to provide adaptive feedback to drivers. Unlike systems that rely on complex in-vehicle sensors, this approach utilizes crowdsensed data to evaluate driving behavior under various contextual conditions.

2.1.1 Methodology

The authors utilized a dataset of crowdsensed telemetric data to engineer several GPS-derived driver risk indicators. These indicators included specific thresholds for harsh braking, rapid acceleration, overspeeding, and turning rates computed directly from bearing changes. Additionally, the system incorporated contextual flags based on nighttime driving and rainy weather conditions. These engineered indicators provided a structured approach for extracting interpretable driving risk features from raw telemetric data. Various machine learning models were then applied to evaluate how these indicators effectively capture driving volatility and predict overall driver risk.

2.1.2 Advantages

The proposed system introduces validated feature engineering methods for transforming raw GPS, temporal, and weather information into meaningful driver behavior indicators. By relying on well-defined, threshold-based rules, the approach provides an easily interpretable and scalable solution for telematics-based risk assessment.

2.1.3 Disadvantages

One major limitation is that the proposed thresholds were validated within a specific machine learning pipeline and regional context. Therefore, their numerical values may require signifi-

cant recalibration when applied to different risk assessment frameworks or different geographical driving environments.

2.2 A New CNN Based Object Detection System for Autonomous Mobile Robots Based on Real World Vehicle Datasets

This paper by Udink Aulia et al., published in Heliyon (Elsevier, 2024), presents a lightweight object detection system designed for autonomous mobile robots. The study addresses the challenge of deploying accurate vehicle detection models on hardware-constrained edge devices. By leveraging transfer learning on a customized dataset, the proposed system aims to balance real-time processing speeds with high detection accuracy in real-world environments.

2.2.1 Methodology

The authors developed a vehicle detection system using the SSD MobileNetV2 FPN-Lite architecture, which is known for its efficiency on mobile and edge devices. To overcome the limitations of generic pre-trained models, they employed transfer learning. The pre-trained detector was fine-tuned on a newly collected and annotated real-world vehicle dataset specifically designed for the target deployment environment. Several training configurations were tested and compared to identify the optimal hyperparameters that maximize the mean Average Precision (mAP) while keeping inference latency low enough for real-time mobile robot navigation.

2.2.2 Advantages

The study demonstrates that transfer learning and fine-tuning of a lightweight SSD MobileNetV2 detector can significantly improve vehicle detection accuracy while maintaining strict real-time performance. This provides a highly practical approach for domain-specific deployment on resource-constrained edge devices.

2.2.3 Disadvantages

Detection performance heavily depends on the quality, size, and diversity of the fine-tuning dataset. Consequently, adapting the model to completely new environments or different weather conditions requires substantial additional data collection and manual annotation

efforts.

2.3 Using Contextual Data to Predict Risky Driving Events: A Novel Methodology from Explainable Artificial Intelligence

This paper by Leandro Masello et al., published in *Accident Analysis & Prevention* (Elsevier, 2023), investigates how contextual factors influence risky driving events. The research introduces a methodology based on explainable artificial intelligence (XAI) to move beyond isolated risk variables. Instead of treating speed or braking in a vacuum, the proposed framework analyzes how combined environmental and traffic contexts compound to create hazardous driving situations.

2.3.1 Methodology

The authors utilized large-scale telematics data combined with contextual driving factors, including varying speed limits, real-time weather conditions, traffic density, road characteristics, and other environmental variables. They trained supervised machine learning models, specifically tree-based ensembles, to predict the occurrence of risky driving events. To interpret the complex interactions between these variables, they applied XAI techniques like SHAP (SHapley Additive exPlanations). This allowed them to quantify the exact contribution of each contextual factor—and their non-linear combinations—toward the likelihood of a risky driving event occurring.

2.3.2 Advantages

The methodology effectively demonstrates that driving risk is determined by the combined, compounded influence of multiple contextual factors rather than isolated variables. It provides a strong, evidence-based foundation for context-aware risk assessment and dynamic feature weighting in driver assistance systems.

2.3.3 Disadvantages

The proposed methodology relies heavily on large-scale historical telematics data for supervised machine learning. As a result, the models require extensive retraining and recalibration when applied to completely different driving environments, varying vehicle types, or new geo-

graphical datasets.

2.4 Optimizing Monocular Driving Assistance for Real Time Processing on Jetson AGX Xavier

This paper by Huy-Hung Nguyen et al., published in IEEE Access (IEEE, 2024), presents a monocular camera-based Advanced Driver Assistance System (ADAS) optimized for edge computing. The study focuses on maximizing the efficiency of vision-based perception algorithms so they can perform complex road scene understanding in real time. The proposed system identifies multiple road users simultaneously without relying on expensive LiDAR or radar sensors.

2.4.1 Methodology

The authors developed a unified vision pipeline that employs a lightweight object detection model to process live monocular camera feeds. The system was specifically optimized and deployed on the NVIDIA Jetson AGX Xavier edge computing platform. They integrated techniques such as tensor optimization and reduced precision inference to accelerate the detection of vehicles, pedestrians, and traffic signs. The performance of the optimized models was then rigorously benchmarked against baseline architectures to evaluate the trade-offs between frame rate, processing latency, and overall detection accuracy in dynamic driving scenarios.

2.4.2 Advantages

The research demonstrates that a single monocular camera, combined with heavily optimized lightweight object detection, can reliably perform real-time perception tasks for driver assistance. This validates the feasibility of camera-based road scene understanding without needing specialized, expensive sensors.

2.4.3 Disadvantages

The system's architecture and performance enhancements are heavily optimized specifically for the Jetson AGX Xavier hardware platform. This makes the deployment strategy and acceleration techniques less directly transferable to other edge platforms or standard smartphone-based implementations.

2.5 Balancing Transparency and Control: The Impact of AI Explanation Detail on User Perception in Automated Vehicles

This paper by Jakob Peintner et al., published in *Transportation Research Interdisciplinary Perspectives* (Elsevier, 2025), investigates the effect of AI explainability on user trust in automated driving systems. The study explores how different levels of detail in safety explanations affect a user's perception of the vehicle's decision-making. The goal is to find the optimal balance between providing enough transparency to build trust without overwhelming the user with excessive information.

2.5.1 Methodology

The authors conducted a comprehensive virtual reality (VR) driving study involving human participants interacting with a simulated Level-4 automated vehicle. They tested four different AI explanation levels: “None” (no explanation), “How” (describing the vehicle's action), “How & Why” (describing the action and the reasoning), and “Contrastive & Counterfactual” (explaining why an alternative action was not taken). Through surveys and behavioral monitoring during the VR simulation, they evaluated the influence of each explanation level on user trust, cognitive load, user experience, and perceived safety.

2.5.2 Advantages

The study provides empirical evidence that concise “How & Why” explanations offer an effective balance between transparency and usability. It establishes a clear, evidence-based approach for designing understandable and trustworthy driver assistance recommendations without causing cognitive overload.

2.5.3 Disadvantages

The findings are strictly based on passenger interactions within a simulated Level-4 highly automated driving environment. Consequently, these results may not directly generalize to real-time, manual driver assistance systems where the human user actively controls the vehicle and requires split-second alerts.

Chapter 3

Requirement Specification

3.1 Functional Requirements

3.1.1 Object Detection

The system shall detect vehicles and obstacles in real time using the smartphone's rear camera and a lightweight object detection model.

3.1.2 Contextual Risk Assessment

The system shall continuously calculate a driving risk score based on combined inputs such as vehicle speed, proximity to detected objects, and environmental conditions.

3.1.3 Transparent Safety Recommendations

The system shall provide clear, explicit reasoning for its risk assessments. The explanations shall help users understand the important factors influencing the current safety recommendations.

3.1.4 Telemetry and GPS Analysis

The system shall use the smartphone's GPS receiver to gather live speed, location, and bearing information. This data shall be used to evaluate driving behavior and compute telemetry-based risk indicators.

3.1.5 Weather and Environmental Monitoring

The system shall obtain current weather information using the OpenWeatherMap API. Weather information, such as rain and visibility, shall be used to adjust risk calculations dynamically.

3.1.6 Real-Time Visual Feedback

The system shall provide an intuitive dashboard interface displaying the live camera feed, current risk status, and safety alerts directly on the mounted mobile device.

3.1.7 Alerts and Notifications

The system shall issue timely, context-aware warnings to the driver regarding unsafe following distances, speed violations, and adverse weather conditions.

3.2 Hardware Requirements

3.2.1 Smartphone Device

A modern smartphone serves as the primary target platform for the application, providing the necessary computational capability, memory, and internet connectivity to run the system entirely on-device without requiring external processing hardware.

3.2.2 Rear Camera

The smartphone's rear camera is used to capture live video feeds of the road scene. The continuous stream of frames is passed to the object detection model to identify vehicles and vulnerable road users in real-time.

3.2.3 GPS Receiver

The built-in GPS receiver of the smartphone is utilized to gather real-time telemetry data. It provides continuous updates on vehicle speed, location coordinates, and bearing, which are critical for driving behavior analysis and risk assessment.

3.2.4 Internet Connectivity

A stable mobile data or Wi-Fi connection is required to retrieve real-time weather information and forecasts through external APIs, ensuring the risk assessment engine operates with up-to-date environmental context.

3.3 Software Requirements

The ContextDrive system utilizes a modern mobile application architecture. It is primarily developed using Dart, Flutter, Google LiteRT, and integrates advanced object detection models and weather services.

3.3.1 Dart

Dart is the primary programming language used for the application. It provides a robust, object-oriented framework for writing both the user interface and the underlying business logic, including the risk assessment engine and telemetry processing.

3.3.2 Flutter

Flutter is used as the cross-platform application framework to develop the user interface. It provides a modular and responsive foundation for features such as the live camera dashboard, risk status displays, and setting configurations, ensuring consistent performance.

3.3.3 Google LiteRT

Google LiteRT (formerly TensorFlow Lite) is employed as the on-device machine learning inference framework. It enables the application to execute complex object detection algorithms directly on the smartphone with low latency and high efficiency.

3.3.4 YOLOv8 Nano (YOLOv8n)

YOLOv8 Nano is the lightweight object detection model integrated into the system. It is specifically optimized for edge devices and is responsible for detecting and tracking vehicles and other obstacles in the camera's field of view in real time.

3.3.5 OpenWeatherMap API

The OpenWeatherMap API is used to fetch current weather conditions based on the user's GPS coordinates. This environmental data provides critical context, such as rain or poor visibility, which is integrated directly into the contextual risk assessment calculations.

Chapter 4

Proposed System and Design

4.1 Proposed System

The proposed system is a smartphone-based platform that combines computer vision, GPS telemetry, and weather intelligence to give drivers reliable and context-aware safety assistance. The system is divided into five modules.

1. **Vision and Object Detection Module:** Captures real-time road scenes using the smartphone camera and detects vehicles and obstacles using a lightweight YOLOv8 Nano model.
2. **Telemetry and GPS Module:** Collects real-time speed, location, and bearing data from the smartphone's built-in GPS receiver to track driving behavior.
3. **Environmental Context Module:** Retrieves live weather and visibility conditions via external APIs based on the current location.
4. **Risk Engine and Reasoning Module:** Combines visual, telemetry, and environmental data to calculate a continuous driving risk score and provides transparent explanations for its assessment.
5. **User Interface Module:** Displays the live camera feed, current risk status, warnings, and safety recommendations through an intuitive dashboard.

The working of the system in brief is as follows. The driver mounts the smartphone on the dashboard and starts the application. The camera begins capturing the road ahead, while the GPS receiver continuously tracks vehicle speed and location. The environmental module simultaneously checks the local weather conditions. The risk engine processes these inputs together—evaluating the distance to detected vehicles, current speed, and weather-related visibility—to determine a driving risk score. If a hazard or unsafe condition is detected, the reasoning module generates a transparent explanation. Finally, the user interface displays the live feed alongside context-aware visual warnings and safety recommendations to assist the driver.

4.2 Feasibility Study

4.2.1 Technical Feasibility

The system uses computer vision algorithms, smartphone sensors, weather APIs, and mobile application technologies, all of which are mature and well documented. It can be developed using open source frameworks like Flutter and Google LiteRT, alongside lightweight object detection models such as YOLOv8 Nano. Since modern smartphones are equipped with high-quality cameras and GPS receivers, the integration of on-device processing without external computing hardware is technically possible. Existing research has already shown that monocular camera-based perception and telematics-based risk assessment work in practice for driver assistance.

4.2.2 Operational Feasibility

The system provides a simple and intuitive dashboard interface for drivers with different technical backgrounds. It delivers real-time risk assessment, visual warnings, and transparent safety recommendations to improve driving decisions. Since the application runs entirely on the driver's smartphone mounted on the dashboard, it requires minimal setup and integrates easily into daily driving routines. The clear, explicit reasoning provided alongside alerts ensures that the system is easily understandable and builds user trust without causing cognitive overload.

4.2.3 Economic Feasibility

The project reduces the likelihood of accidents through proactive risk assessment and timely safety warnings. The development cost is low because open source software frameworks and freely available AI models are used. Unlike traditional Advanced Driver Assistance Systems (ADAS) that require expensive, specialized vehicle hardware, this system leverages the smartphone the user already owns. The minimal cost associated with standard mobile data usage for weather APIs makes the system highly affordable. The expected improvement in driver safety and accident prevention is significantly greater than the minimal operational cost of the system.

4.3 Design

The design of the system is user centric and modular. The main design diagrams are explained below.

3. **Context Understanding Layer:** This central layer consolidates diverse data points to create a comprehensive understanding of the current driving environment and situation. The context understanding engine builds a high-level situational model by receiving the location stream, weather data, timestamp, and analyzed features, ultimately generating a critical data stream called the driving context.
4. **Decision Layer:** This layer analyzes the comprehensive driving context to assess risk and formulate advice in two connected stages:
 - *Driving Risk Assessment:* Evaluates the context data for potential hazards and risk levels, generating a risk analysis stream.
 - *Recommendation & Explanation Engine:* Generates actionable advice and the rationale behind that advice based on the risk analysis, resulting in a consolidated assessment and recommendation stream.
5. **Output Layer:** This final layer presents the system's findings and advice to the user. It receives the high-level driving context and the assessment and recommendation data. The presentation modules include the Risk Level, Safety Alert, and Recommendation.

4.3.2 Use Case Diagram

The use case diagram illustrates the main functionalities of the ContextDrive system and the interactions of its primary actor, the driver. The driver can start a trip, view the live camera dashboard, and receive real-time driving risk assessments based on speed and proximity. The driver can also review the transparent safety explanations generated by the system to understand the factors influencing the current warnings.

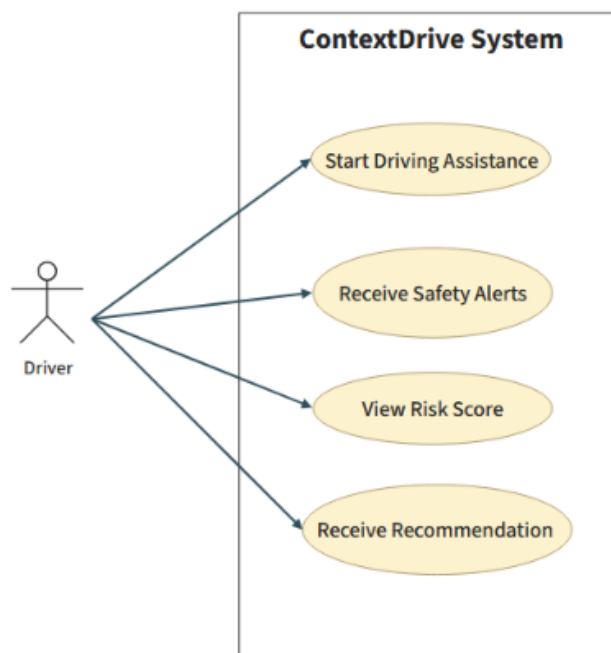


Figure 4.2: Use Case Diagram

4.3.3 Level 0 DFD

Level 0 represents the complete ContextDrive system as a single process, as shown in Figure 4.3. It illustrates the high-level flow of information:

- **Driving Context Data (Input):** Raw environmental and sensor data (location, weather, time, camera) fed into the system.
- **ContextDrive (Processing):** The core engine that receives the raw data, analyzes the driving situation, and evaluates potential risks.
- **Risk Alerts & Driving Recommendations (Output):** The final result generated by ContextDrive, delivering safety warnings and driving advice to the user.



Figure 4.3: Data Flow Diagram - Level 0

4.3.4 Level 1 DFD

Level 1 decomposes the ContextDrive system into major processes, as shown in Figure 4.4. The flow is structured as follows:

- **Inputs:** Four external data sources feed into the central system:
 - *Smartphone Camera:* Sends Camera Frames.
 - *Smartphone GPS:* Sends Speed / Location.
 - *Weather API:* Sends Weather Data.
 - *Smartphone Clock:* Sends Time & Date.
- **ContextDrive System:** The central processing hub that receives and combines all four input streams.
- **Driver:** The final recipient who receives the Risk Score, Safety Alerts, and Recommendations generated by the ContextDrive System.

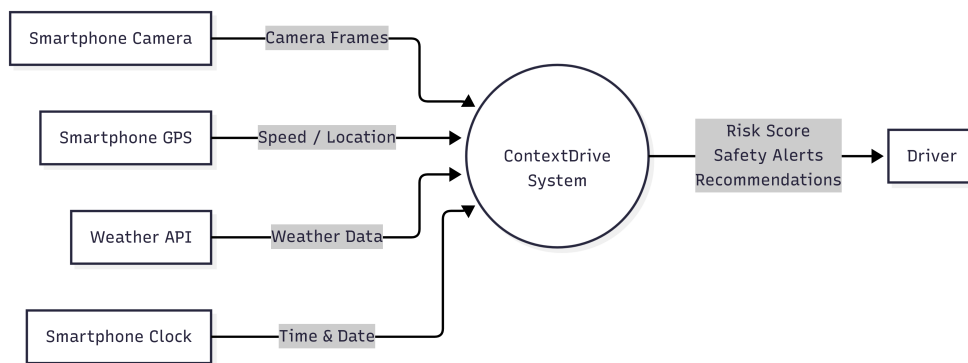


Figure 4.4: Data Flow Diagram - Level 1

4.3.5 Level 2 DFD

Level 2 DFD provides a more detailed view of the internal sub-processes within the core modules of the system, as shown in Figure 4.5.

- **Inputs:**
 - *Smartphone Camera:* Sends Camera Frames to the Vehicle Detection process.
 - *Smartphone GPS:* Sends Speed/Location to the Context Understanding process.
 - *Weather API:* Sends Weather Data to the Context Understanding process.
 - *Smartphone Clock:* Sends Time & Date to the Context Understanding process.

- **Processes:**

- *Vehicle Detection*: Receives Camera Frames and outputs Vehicle Information to the Context Understanding process.
- *Context Understanding*: Combines Vehicle Information, Speed/Location, Weather Data, and Time & Date to output Driving Context to the Driving Risk Assessment process.
- *Driving Risk Assessment*: Analyzes the Driving Context and sends a Risk Score to the Recommendation & Explanation process.
- *Recommendation & Explanation*: Takes the Risk Score and sends the final Alert and Recommendation to the Driver.

- **Output:**

- *Driver*: The final recipient who receives the Alert and Recommendation from the Recommendation & Explanation process.

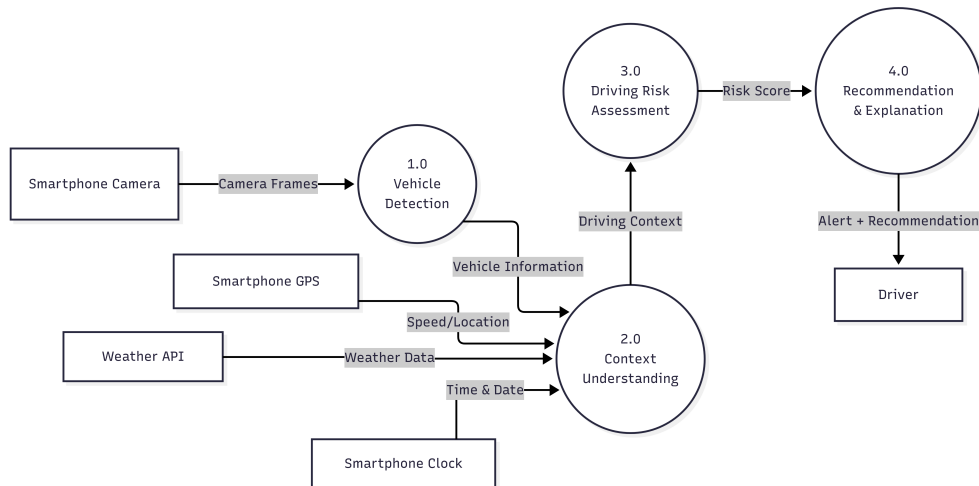


Figure 4.5: Data Flow Diagram - Level 2

4.3.6 Methods and Techniques

The ContextDrive system integrates various methods and techniques to support drivers in making informed, safe decisions. The system combines computer vision-based object detection, GPS telemetry tracking, weather data analysis, deterministic risk assessment, and transparent explainability mechanisms. These techniques work together to provide personalized and data-driven safety recommendations.

Computer Vision-Based Object Detection

The system uses the YOLOv8 Nano model via Google LiteRT to detect vehicles and obstacles

in the live camera feed. This lightweight model processes camera frames efficiently on-device, providing bounding boxes and proximity estimations for critical road elements.

GPS Telemetry Tracking

The system collects real-time speed, location, and bearing data from the smartphone's built-in GPS receiver. This telemetry data is used to continuously monitor the vehicle's dynamic state and movement patterns.

Weather Data Analysis

Weather and visibility information is obtained using the OpenWeatherMap API based on current GPS coordinates. Parameters such as visibility distance, precipitation, and extreme weather alerts are incorporated to adjust the risk sensitivity of the system.

Deterministic Risk Assessment

The system employs a rule-based deterministic risk engine to calculate a continuous driving risk score. It evaluates multiple factors, including vehicle speed, proximity to detected objects, and adverse weather conditions, to classify the driving situation into distinct risk levels.

Transparent Explainability

To build trust and prevent cognitive overload, the system generates clear, explicit explanations alongside its warnings. This ensures the driver understands exactly which environmental or behavioral factors contributed to the current safety alert.

Real-Time Driver Interface

The application provides a responsive Flutter-based dashboard that overlays the risk score, safety alerts, and actionable recommendations directly onto the live camera feed, ensuring seamless integration into the driving experience.

Chapter 5

Progress in Project

5.1 Implementation

The ContextDrive system has been implemented as an integrated advanced driver-assistance platform combining edge-based computer vision, GPS telemetry, environmental intelligence, and deterministic risk assessment. The implementation connects these diverse modules so that real-time driving data can be processed on-device and presented to the driver through a unified, smartphone-based interface.

5.1.1 Module Split-up

Data Acquisition and Pre-processing Module

The data acquisition module collects the dynamic driving parameters required by the system. The implemented setup uses the smartphone's built-in sensors to obtain real-time video frames from the rear camera and speed, location, and bearing data from the GPS receiver. Weather information is obtained through an external weather service to provide environmental context such as visibility and precipitation. The collected values are validated and synchronized before being passed to the processing modules, ensuring that the input data is robust for risk evaluation.

Vision and Object Detection Module

The vision module is responsible for identifying potential hazards on the road ahead. The implemented model uses a lightweight YOLOv8 Nano architecture optimized for mobile devices. It processes the live camera frames to detect vehicles and obstacles, generating bounding boxes and estimating proximity. The continuous object detection results are used by the risk assessment module to determine safe following distances and potential collision risks.

Environmental Context Module

The environmental context module provides critical situational awareness based on the vehicle's current location. The system queries the OpenWeatherMap API to retrieve real-time weather conditions. By considering factors like fog, heavy rain, or low visibility, this module provides contextual weights that dynamically adjust the sensitivity of the risk assessment algorithms, converting raw weather data into actionable driving intelligence.

Risk Assessment Module

The risk assessment module evaluates the integrated driving data to compute a continuous driv-

ing risk score. The system employs a deterministic, rule-based engine that compares vehicle speed, object proximity, and weather conditions against predefined safety thresholds. Instead of merely logging data, the module classifies the current driving situation into distinct risk levels (e.g., Safe, Warning, Critical) to facilitate immediate driver response.

Transparent Explainability Module

To make the automated risk assessments easier to understand and trust, an explainability module generates explicit reasoning for its decisions. Whenever a warning or critical alert is triggered, the system analyzes the contributing factors (e.g., “High speed in low visibility” or “Following too closely”). This approach ensures that the driver understands exactly why an alert was issued, reducing cognitive overload and improving the interpretability of the assistance system.

User Interface and Dashboard Module

The user interface module provides a seamless and intuitive dashboard for the driver. The mobile application overlays the live camera feed with real-time bounding boxes, the current risk score, safety alerts, and transparent recommendations. It is designed to deliver critical information efficiently without distracting the driver from the road.

5.1.2 Tools and Techniques

Frontend Development

The frontend of the system is developed using the Flutter framework with the Dart programming language. Flutter provides a highly responsive, cross-platform UI toolkit that enables the creation of a smooth, real-time camera dashboard. It handles the display of live video feeds, floating UI overlays for risk scores, and the presentation of transparent safety explanations.

Edge AI and Computer Vision

The core perception capabilities are powered by Google LiteRT (formerly TensorFlow Lite), which allows the YOLOv8 Nano model to execute efficiently directly on the smartphone hardware. This edge-AI approach ensures that video frames are processed locally in real-time, eliminating network latency and preserving driver privacy since video data never leaves the device.

Telemetry and Location Services

The system utilizes native smartphone GPS sensors accessed via Flutter location and geolocator packages. This provides the continuous stream of speed, heading, and coordinate data necessary for the risk engine to accurately monitor driving dynamics.

Environmental Data Services

The OpenWeatherMap API is used as the external cloud service for retrieving hyper-local weather data. Standard HTTP request libraries in Dart handle the API communication, fetching

relevant environmental context based on the current GPS coordinates to enhance the system's situational awareness.

Chapter 6

Expected Outcome

The expected outcome of the ContextDrive project is a user-friendly, smartphone-based advanced driver-assistance system (ADAS) that assists drivers in making informed, safe decisions using real-time camera, GPS, and environmental information. The system integrates edge-based computer vision, continuous telemetry monitoring, environmental intelligence, deterministic risk assessment, and transparent explainability into a single mobile application. The vision and object detection module is expected to identify potential hazards, such as leading vehicles, by efficiently processing live camera frames on-device. The risk assessment module is expected to provide practical safety evaluations by analyzing vehicle speed, object proximity, and adverse weather conditions, helping drivers maintain safe following distances and adapt to changing environments. The transparent explainability module is expected to improve trust and interpretability by explicitly identifying the critical factors (e.g., high speed in low visibility) that triggered a safety warning. Overall, the project is expected to provide an integrated smart driving platform that supports data-driven risk evaluation, enhanced situational awareness, and context-aware safety guidance.

The system is also expected to provide a seamless and accessible interface through which drivers can view their live camera feed, real-time risk scores, bounding boxes for detected hazards, and actionable safety recommendations. By bringing these functionalities together in a single smartphone application without the need for expensive dedicated hardware, the project aims to reduce the complexity of driving safety systems and support drivers in adopting safer, more proactive driving habits.

Chapter 7

Conclusion

The ContextDrive project provides an integrated advanced driver-assistance platform that combines edge-based computer vision, GPS telemetry tracking, environmental intelligence, deterministic risk assessment, and transparent explainability. By using continuous vehicle dynamics and environmental data, the system helps drivers identify potential road hazards and receive real-time, actionable safety guidance. The integration of explainable AI improves the transparency of risk assessments, while the edge-processing architecture supports privacy and rapid response times. The mobile-first approach further provides accessible safety features without requiring expensive hardware modifications. Overall, the project aims to support drivers in making informed, data-driven decisions and adopting safer, more proactive driving habits.

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